Curating Real-World Manufacturing Evaluation Datasets to Assess the AI Models and Agentic Harnesses behind Overseer

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Background and problem statement

Overseer is an agentic AI solution built on top of manufacturing stacks, now piloting with industry partners. We need to evaluate both the AI models and the agentic harnesses that run them on real factory processes and catch regressions as we extend Overseer to broader use cases for more partners. However, we do not have purpose-built evaluation datasets for this, and neither are there publicly available ones, so we cannot be confident that the system generalises safely or remains reliable as it evolves.

Aim

Design and build a set of purpose-built evaluation datasets, derived from real manufacturing partner data, that rigorously test Overseer's AI models and agentic harnesses across representative manufacturing tasks and catch regressions as the system expands to broader use cases and more partners, extending the team's existing model-evaluation harness.

Objectives

- Analyse, clean, and structure real manufacturing partner data into task-specific evaluation datasets with clear, checkable ground truth.
- Design evaluation methods and scoring criteria appropriate to each dataset, favouring deterministic, checkable success criteria over subjective judgement.
- Extend the existing harness to support the new manufacturing-specific datasets.
- Evaluate a range of open-weight models and the agentic harnesses/pipelines that run them on real factory processes, against the curated datasets, reporting on comparative performance and suitability.
- Structure the datasets so they can double as a regression check for Overseer.

Expected Outcomes

- A set of purpose-built manufacturing evaluation datasets, curated from real industry data, ready for ongoing use by the team.
- An extended harness capable of running the new evaluation datasets end to end.
- A comparative evaluation report on open-weight and closed-weight models and agentic-harness performance on real manufacturing tasks.

How will this opportunity benefit you?

Skill development in the following areas:

- Systematic evaluation methodology, designing rigorous, checkable ways to assess AI systems and agentic pipelines.
- Data curation and dataset design for AI evaluation, working with real-world, messy industrial data.
- Data analytics, analysing and interpreting real manufacturing data to identify meaningful evaluation scenarios
- Software engineering practice, extending an existing evaluation harness codebase.

Other details

- Located on-campus at Burwood
- 10 weeks in duration (Unpaid)
- Undertaken as 2 days per week for a total of ~160 hours
- To be completed within the 12-week trimester period